
O-RAN Work Group 1 (Use Cases and Overall Architecture)

RIC Enabling NTN Deployments

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Foreword

This Technical Report (TR) has been produced by WG1 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Introduction

Support for Non-Terrestrial Networks (NTN) has been introduced in the 3GPP specifications starting from Release 17. Release 18 has been recently completed, with further enhancements in development as part of Release 19 and expected in the upcoming Release 20.

NTN has gained substantial momentum in the telecommunication industry, with the first services in support of IoT and direct-to-device messaging launched, and more deployments including NR-NTN upcoming. NTN presents an emerging opportunity for Open RAN and there are multiple research/development projects and evaluations in progress focusing on NTN implementation leveraging Open RAN technologies. Non-Terrestrial Networks, in particular NGSO constellations, dynamic GSO, as well as hybrid multi-orbit constellations, present complex architectural, control, radio resource management, mobility and orchestration challenges which could be addressed by Open RAN technologies.

The O-RAN ALLIANCE is currently developing specifications for O-RAN RIC-enabled optimization and autonomous management of terrestrial networks. Standardization of O-RAN RIC-enabled optimization and autonomous management for NTN and converged NTN+TN interworking deployment scenarios is essential for the NTN operator constellation planning and architectural phases, as a key enabler for their commercial success.

This intelligent management of the RAN and its operation is naturally addressed by the Non- and Near-RT RICs. However, while there are many use cases documented for their employment in terrestrial 5G networks, moving the network functions into space brings in a whole new set of challenges and constraints, and opens up completely different novel implementation approaches for the existing O-RAN use cases (such as TS and QoS optimization), as well as a new host of use cases that are specific to NTN. For example to account for the new control signaling features introduced for NTN in 3GPP Releases 17 and 18 (such as SIB19 and timer/distance-based CHO), as well as to address challenges unique to NTNs.

This technical report investigates use cases applicable to NTN and presents a gap analysis on possible impacts to WG2 and WG3 specifications of the O-RAN ALLIANCE, in preparation for normative work in WG2 and WG3. Two main groups of use cases are identified:

- **Use Case Group #1:** How the TN and NTN segments will intelligently cooperate and provide a seamless experience to the users as they traverse between TN and NTN, and vice-versa, while making sure that the available spectral resources are optimally utilized.
- **Use Case Group #2:** Once in the NTN segment, how are the users and their traffic flows going to be intelligently managed by the SNOs such that they are assuring the types of services being sold while accounting for temporospatial variations in motion of the space and aerial assets, weather, spectrum availability, onboard power and hardware constraints, different terminal classes, fast mobility, Line of Sight (LoS), QoS, etc.

1 Scope

The present document specifies a technical report on RIC enabling NTN deployments use cases.

2 References

2.1 Normative references

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] Non-Terrestrial Networks (NTN)

NOTE: Available at: <https://www.3gpp.org/technologies/ntn-overview>

[i.2] 3GPP TR 38.821: “Solutions for NR to support Non-Terrestrial Networks (NTN)”

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

CHO	Conditional Handover
EFC	Earth Fixed Cell
EMC	Earth Moving Cell
FDD	Frequency Division Duplex
FR1	Frequency Range 1
GSO	Non-Geostationary Orbit
HO	Handover

IoT-NTN	Internet of Things Non-Terrestrial Networking
ISL	Inter-Satellite Link
KPI	Key Performance Indicator
LEO	Low Earth Orbit
LoS	Line of Sight
NB-IoT	Narrow Band Internet of Things
NR-NTN	New Radio Non-Terrestrial Networking
NTN	Non-Terrestrial Network
NGSO	Non-Geostationary Orbit
OTA	Over-The-Air
PCI	Physical Cell Identity
QEFC	Quasi-Earth-Fixed Cell
RIC	Radio Intelligent Controller
SMTTC	SSB Measurement Timing Configuration
SNO	Satellite Network Operator
SSB	Synchronization Signal Block
TDD	Time Division Duplex
TN	Terrestrial Network
UAV	Unmanned Aerial Vehicle
UE	User Equipment
UML	Unified Modeling Language

4 NTN development status in standard organizations

4.1 3GPP

4.1.1 Overview

Non-Terrestrial Networks (NTN) are a significant focus within the 3GPP standards, particularly for integrating satellite into 5G and beyond. Here's a summary of the current development status:

1. Release 15 and 16: Study phase on scenarios, use cases, channel models and impact of NTN to the 5G System, as well as solutions for 5G New Radio (NR) support for Non-Terrestrial Networks.
2. Release 17 and 18: These releases introduced NTN with a transparent (bent-pipe) architecture, where satellites act as radio repeaters without on-board processing capabilities. This approach was chosen for its simplicity and early deployment potential. Both NR-NTN and IoT-NTN are supported. Release 18 introduced support for NR NTN above 10 GHz, such as broadband in Ka band.
3. Release 19: This release marks a major milestone with the introduction of a regenerative payload architecture. In this scenario, a complete gNB (Next Generation Node B) is placed on the satellite, allowing for greater integration and flexibility, and opening up new constellation topologies and use cases). This architecture supports packet processing directly on the satellite, enabling features like improved mobility management.
4. Release 20: The next release is currently in the planning phase and a 3GPP RAN workshop in RAN#106 (12/2024) discussed potential NTN features for Rel-20. The initial work plan for Rel-20 was decided in RAN#108 (06/2025).

Technological Challenges: The development of NTN involves unique challenges such as high Doppler shifts, longer propagation delays, and larger cell sizes due to the high altitudes of satellites. Enhanced beam management and regulatory considerations are also crucial. Most of these challenges are already covered by Rel-17 and 18.

Future Prospects: The ongoing work aims to further integrate NTN with terrestrial networks, enhancing global coverage and resilience. The regenerative payload architecture is expected to support advanced functionalities like "store and forward," which allows satellites to store data temporarily and forward it when connectivity is restored, and also UE-satellite-UE communication with reduced latency.

4.1.2 R17/R18/R19 work items for NTN

Release 17:

1. Transparent payload architecture with FDD systems.
2. Adaptation to physical and access layer aspects, radio access network, system architecture, radio resource management, and RF requirements.
3. Support for NTN-NR (5G New Radio (NR)) based satellite access in FR1 bands for handheld devices.
4. Support for NTN-IoT (NB-IoT and eMTC) based satellite access for massive IoT and handheld messaging use cases.

Release 18:

1. Enhancements for NR-NTN and IoT-NTN, especially the time and location based conditional handover (CHO) for mobility enhancements.
2. Enablers for NR based satellite access in bands above 10 GHz (including introduction of Ka-band, see 3GPP TS 38.108 and 38.101-5).
3. Support for fixed and moving platforms (e.g., aircraft, vessels, UAVs) and building-mounted devices.

Release 19

1. Downlink Coverage Enhancements with CD-SSB periodicity relaxation and enhanced beam-hopping support
2. Uplink capacity enhancements.
3. Support for a non-terrestrial network architecture with 5G system functions on board the NTN vehicle (regenerative payloads), including the possibility for inter-satellite links (ISL).
4. Use of RedCap devices within FR1 NTN.
5. Store & Forward (S&F) operation based on regenerative payload to support non-complete constellations.
6. Specification of Ku-band for NTN operations
7. Support for half duplex operation in TDD and FDD scenarios for IoT-NTN
8. New L- and S- frequency band definitions
9. Over-The-Air (OTA) antenna requirements and performance framework for NTN

4.2 O-RAN

4.2.1 Overview

To the group's best knowledge, no approved contributions or normative work on NTN exist so far in O-RAN ALLIANCE, but there have been some requests for the definition of new WIs and research areas in nGRG.

4.2.2 Study and work items for NTN

The following proposals have been submitted so far:

1. New feature work for NTN deployments enabled by RIC.
2. Study on open TN-NTN architecture based on O-RAN
3. Research for requirements on O-RAN based satellite clustering for NTN in nGRG
4. Work for regenerative NTN support in O-RAN in WG1 UCTG

5 Key NTN use cases analysis in O-RAN enabled by RIC

5.1 Use Case Group 1: TN and NTN interworking and cooperation

5.1.1 Overview

In most cases, these NTN networks and services are not anticipated to be completely standalone offerings, but rather intended at least in part as a complement to Terrestrial Networks, for example to provide coverage extension in white spots and hard to reach areas, 3D coverage at altitude, or a layer of resiliency. In fact, the ability for NTN to eliminate or alleviate coverage gaps in the Terrestrial Network was one of the key motivations for the introduction of NTN in 3GPP since the beginning.

Based on the current market trends, it appears NTN deployment motivations will in most cases fall into one of the following categories:

- Existing Satellite Network Operators (SNOs) who are looking to serve the NTN market in addition to or in evolution of their existing non-3GPP satellite services.
- New entrant SNOs who are seeking to uniquely provide NTN services.
- Mobile Network Operators (MNOs) who are seeking to extend their network via NTN.
- New entrant MNOs who are seeking provide homogenous TN and NTN coverage

Therefore, it is expected that many MNOs may seek to extend their own Terrestrial Network services via NTN, however the current market trends indicate that they will likely not seek to deploy and operate their own satellites, but rather form agreements with one or multiple NTN SNOs, who will provide coverage over areas of interest on behalf of their partner MNOs.

For this to be achieved, the MNO's and SNO's networks will need to interact and coordinate with one another at some level. The required degree of interaction and information sharing required between the two operator domains will depend on the requirements of the specific use case and on the depth of integration between the two segments. This interaction is essential in order to for example make the NTN network aware of where the boundaries of the terrestrial network exist, share handover preparation information, reselection targets, etc. and conversely make the terrestrial network aware of when an NTN is available to its users.

The following clauses describe some of the use cases that fall within the category of TN and NTN interworking and cooperation and that have been grouped under Use Case Group 1 (UCG#1).

5.1.2 Use Case 1-1: TN-NTN Coverage Information Sharing

5.1.2.1 Use Case Description

In order for an MNO and their partner SNO(s) to be able to support efficient complementary coverage via NTN, the TN network may need to be aware of the available NTN coverage footprint, and vice versa. This can include current coverage (i.e. already active) as well as potential coverage (i.e. not yet active, but possible), and will need to include some details of the coverage to be provided (e.g. applicable NTN cells, their configuration and corresponding serving satellite ephemeris).

Depending on a number of operational factors and system constraints, such as feeder link availability, power and radio resource availability of the satellites forming the NTN constellation, the active cells, allocated resources (and thus capacity), as well as the specific coverage areas that an NTN operator chooses to serve or not serve may change over time. In addition, due to orbital precession, the specific satellite(s) serving a given cell coverage area on the ground at any point of time (e.g. in a scenario with Earth-fixed or quasi-Earth-fixed Cells) also change over time. Thus there is a need for the NTN segment to be able to communicate this information to the TN segment, such that the TN segment can be aware of which NTN cells are available or might become available in the future, as well as update the information being broadcast in SIB19/SIB31 messages in terrestrial edge cells.

Likewise, in the reverse direction, an NTN operator may normally not be aware of the footprint, topology, configuration or even presence of terrestrial networks within the coverage of an NTN, unless this information has been separately sourced or disclosed by the MNO. Whilst some of this information can be shared at network planning by the TN MNO to the NTN SNO,

the TN MNO network is expected to evolve over time, potentially also dynamically via the introduction of network energy saving (NES) techniques, and thus may need to be updated at runtime.

In scenarios where the NTN coverage may totally or partially overlap with the TN coverage or its edge, by having awareness of the TN footprint or of its edge, the NTN operator may be able to selectively deactivate NTN coverage in overlapping areas, in order to save system power, capacity, or to enable coverage in other locations. In scenarios with no overlap (e.g. due to TN spectrum reuse in space), the NTN operator may need to adjust the NTN coverage edge to match as closely as possible the TN coverage edge, at least for certain frequencies.

5.1.2.2 Problem statement and challenges

In order to fulfil the requirements of these scenarios, the TN and NTN segments would need to have a method of exchanging at least the minimum necessary information on their respective coverage footprints, required to fulfil complementary coverage functionalities. The current 3GPP- and O-RAN-specified interfaces between operator domains, such as the roaming interfaces and other interfaces at the Core network and RAN level (e.g. Xn), do not support sharing of this type of information.

Sharing of neighbor cell information, including neighbor relationships, and other coverage related information within a single network domain (e.g. within a TN or an NTN) is typically expected at least in part to be addressed by interfaces such as X2/Xn. However, in the interaction between TN and NTN this is either not sufficient (e.g. due to lack of the necessary primitives), or outright not possible, such as in the scenario where SNO and MNO, with both networks assumed to be aligned with 3GPP and O-RAN architecture, are two distinct operator domains, where inter-operator Xn is currently not contemplated (or implemented) and presents potential feasibility issues. In addition, both the TN footprint, topology and configuration, as well as the NTN footprint, topology and configuration, are expected to be continuously evolving and dynamic in nature, especially as operators (MNOs and SNOs alike) increasingly target maximization of energy and deployment efficiencies. Therefore, traditional manual or static mechanisms of sharing this information are not going to be fit for purpose, and there is instead a need for automated mechanisms that can dynamically provide updates on the changes to the coverage in a timely manner (order of seconds at least), as soon as they happen. It is also important that this sharing of information is performed securely and with a fine-grained level of control on the type and amount of information that is respectively shared, at any given instant of time.

Therefore, there is a gap and an identified need for a secure, programmatic and reactive mechanism for TN and NTN networks in different operator domains to interface and interact, for the purpose of selectively sharing coverage related information.

5.1.3 Use Case 1-2: TN-NTN Mobility and Service Continuity

5.1.3.1 Use Case Description

This use case covers the support for mobility and service continuity between a terrestrial network (TN) and a Non-Terrestrial Network (NTN). One of the key value propositions for Non-Terrestrial Networks is to enable complementary coverage for terrestrial networks, and to help MNOs fulfil coverage gaps. As such, enabling or facilitating mobility and service continuity between the TN and the NTN and vice-versa is a key requirement.

Whilst 3GPP defines the procedures at RAN level for TN-NTN idle and connected mode mobility and stipulates the need for sharing of coverage and satellite assistance information (e.g. Ephemeris), the 3GPP specifications do not explicitly address how this information is shared, and further how these procedures are coordinated when the TN and NTN are in different operator domains. This use case thus describes the requirements, expected procedure and possible solution to address these remaining aspects in respect to TN-NTN mobility and service continuity.

In order to be able to support TN-NTN mobility, the TN RAN needs to be aware of the NTN coverage footprint, and specifically which satellite(s) are or will be serving a given coverage area at any point in time, and which are or may be the applicable neighbor NTN cells, with their respective configurations. In 3GPP NTN Rel-17+ systems, some of this information, alongside other details such as Ephemeris of the satellites, is expected as per the 3GPP specifications to be delivered via SIB19 or SIB31 into the TN cells bordering or overlapping with the NTN coverage area. However, in order for the information to be delivered to the TN cells, the terrestrial network has to first be aware of the information in the first place. This information may in many cases need to come from an external source to the MNO network (e.g. an SNO network). In addition, there may be another scenario in which an MNO may wish to provide

Additionally, when it comes to evaluating the candidate handover or reselection opportunities to an NTN cell from a TN cell, there may be a subset of satellites which are better suited as target candidates, and there is a need to study how the O-RAN RIC/SMO architecture might be leveraged in optimizing the selection, for example by tailoring the SMTC list and the NTN-

NeighCellConfigList provided in SIB19 to include only the satellite candidates that actually make for good handover or reselection targets, in some cases based on the specific UE service requirements on RAN and line-of-sight (LoS) conditions.

5.1.3.2 Problem statement and challenges

Whilst 3GPP specifies mechanisms for TN-NTN mobility and service continuity, many of the capabilities and information exchanges required to implement said mobility, and to more in general facilitate the seamless interworking between TN and NTN are currently not addressed by 3GPP specifications, and could on the other hand be effectively addressed by O-RAN.

In order to support mobility from TN to NTN, the TN network must be aware at least of the following information, and their applicability in time:

- Satellite assistance information (e.g. Ephemeris) of the target NTN systems, satellites and cells available for handover or reselection
- Cell topology and availability of the NTN cells neighboring or overlapping the TN coverage edge, that could be candidate target cells for measurement, reselection or handover
- Cell configuration of the NTN cells neighboring or overlapping the TN coverage edge, necessary for measurement configurations, reselection policy configuration and handover triggering

In addition, the network-initiated Connected Mode handovers must also be explicitly triggered by the network, which must be able to:

- Request measurements to the target NTN network, to identify HO cell candidates
- Request HO triggering to the target gNB

In a terrestrial network scenario or an intra-operator TN-NTN scenario, many of these capabilities are or can be enabled by the X2/Xn interface. However, X2/Xn may not be feasible in an inter-domain/inter-operator environment.

In addition, given that between two operators, the network topology might be partially or completely obfuscated, there is no way for the Source gNB to even identify which target gNB endpoint (e.g. gNB-CU) to address the measurement and handover requests, as the X2/Xn interface exists between individual eNBs/gNBs.

Thus, there would be no mechanisms for signaling of neighbour relationships, including for measurement configurations. Furthermore, the TN may not even be aware of the neighbour NTN cells (esp. for candidate cells that are not active, in systems deploying selective coverage and/or beam hopping).

Therefore, these challenges arise with this scenario and the lack of a suitable interface, for example:

- No awareness by the TN of the target NTN cell configuration (e.g. frequencies, PCI, etc) required to configure measurements, HO triggers, Reselection policies.
- No awareness of the gNB topology in the target network (either TN or NTN), and thus no awareness of which gNB to even establish an Xn connection or trigger an HO request or measurement request.
- With 3GPP NTN R17+, the TN might not be aware of Satellite Ephemeris (SIB19/SIB31 content), required by the UE to perform UL pre-compensation
- The NTN coverage may not be active (or may be configured with insufficient duty cycle/capacity) in the target area that the UEs are about to move to (satellite coverage is limited and must be selective)
- The NTN may not have any awareness of incoming load, QoS requirements or UE capabilities
- In 3GPP R19 NTN, SSB Periodicity could be relaxed (e.g. up to 160ms). This can impact UE sync performance/reliability and may need specific SMTC configuration for measurement before TN to NTN handover
- UE capability signaling is very heavy. Forcing the NTN to perform UE capability exchange from scratch may have adverse impact on NTN capacity
- Letting the UE/network declare RLF and Detach the UE, would lead to a network selection/access from scratch, which creates service interruption and unnecessary signalling load.
- The NTN network may not be aware of which (or any) UE in TN are about to be handed over (or reselect to) NTN

- In the reverse direction (NTN to TN), no awareness by the NTN of the target TN cell configuration (e.g. frequencies, PCI, etc) required to configure measurements, HO triggers, Reselection policies

5.1.4 Use Case 1-3: Request of NTN coverage for Disaster Response

5.1.4.1 Use Case Description

In some cases a Terrestrial MNO might wish to explicitly request NTN coverage in a given area, either to fulfil a coverage gap, or to respond to a total or partial outage of the terrestrial network, such as in a disaster scenario. This same capability could also be used simply to request the NTN to allocate additional capacity in a given area, in order to respond to an expected or witnessed increase in demand.

Scenario 1) TN MNO1 identifies a coverage gap in its coverage, or is interested in providing additional coverage overlay, and thus requests NTN coverage to specific target geographical area to NTN SNO1.

Scenario 2) TN MNO1 already has active NTN coverage by NTN SNO1 in a given target geographical area, but identifies a need to allocate additional capacity (e.g. due to an upcoming event).

Scenario 3) During a disaster scenario (e.g. a hurricane), cell towers of TN MNO1 in a given area are affected by an outage (e.g. power outage) and are thus unavailable. TN MNO1 requests NTN coverage (in dedicated or shared spectrum) to the affected target geographical area to NTN SNO1.

The scenarios from 1 to 3 are applicable also when an MNO has a commercial relationship or wishes to establish a commercial relationship with more than one NTN SNO partner.

In addition to the basic coverage or capacity request, a TN MNO may wish to communicate additional information to their NTN SNO partner, for example:

- Expected load or last-known load
- Required capacity
- QoS information and/or Slice policies for specific services that may be required (e.g. public safety, emergency messaging and/or calling)
- List of UEs for prioritization/QoS policy
- Pre-sharing of UE capabilities mapped to explicit list of UEs
- Interference thresholds and coverage edge contours or other constraints (e.g. to prevent interference, in case of terrestrial spectrum reuse)
- Desired cell configuration parameters
- In case of terrestrial spectrum reuse or other form of spectrum lease, information on the spectrum available for deployment and any configuration requirements.

5.1.4.1 Problem Statement

Current methods for an MNO to request NTN coverage from an SNO are entirely reliant on traditional office IT communication (e.g. emails, meetings), and require extensive discussions and pre-agreements not only at the commercial level, but also at the technical level, as a pre-requisite.

With current methods, derivation of technical parameters for requests of either complementary coverage or capacity rely on extensive offline simulations performed by both parties respectively. These simulations are usually based on abstracted knowledge of the network, and relatively limited advance knowledge of the respective systems, also due to confidentiality constraints. From these simulations and plans, a notional NTN network plan is usually derived and agreed by both parties, via extensive discussion. In some cases, the simulations used to derive the plan may not even be fully reflective of the up-to-date conditions of the network (mostly because they are generated not from the live network, but from offline system-level simulator toolsets), nor of the real-world practical conditions in which the system will operate.

In addition, any subsequent update usually needs to go through a similar formal cycle of discussions, simulations, agreements and formal approvals, de-facto preventing the enactment of any reactivity to changing requirements or events in responsive timeframes. In some cases, due to the relatively “static” nature of the plan and of the need to pre-plan in advance, often with a high level of detail, this can also lead to inflexibility in any regulatory filings submitted, where exact static values may be specified or filed for, and thus require an update to the regulatory filings in case some parameters are altered or optimized.

As a consequence, the resulting plan exhibits a few unfortunate byproducts:

- 1) **Staleness:** The NTN network plan derived is relatively static and represents a snapshot of the respective systems at a given moment of time, which may become stale over time, due to being unable to automatically reflect any subsequent changes or optimizations in either the MNO terrestrial network coverage, or the SNO's NTN coverage.
- 2) **Immutability and lack of reactivity:** Changes in the NTN network plan are difficult and slow to introduce, and not only it cannot easily accommodate either changes in the actual respective network topologies, capabilities and plans, but it is also unable to be adapted in responsive time frames, to react to changing or unforeseen needs by either SNO or MNO, such as need to provide Disaster Coverage in response to an outage, spikes in demand or other time-sensitive events.
- 3) **Poor accuracy:** Due to the simulation environments used to derive the technical parameters of the plan, and due to issues with confidentiality, the plan may not reflect the accurate and up-to-date state, capabilities or parameters of the live network, reducing the accuracy of representation. Its accuracy also to the physical live environment and its interaction with terrain, weather and other factors is also very limited.
- 4) **Conservativeness and sub-optimality:** Due to the above issues, the plans are usually defined with relatively (sometimes extremely) conservative margins, in order to avoid, or at least reduce, any risks of disruption to user experience. Inevitably, this also results in significant sub-optimality.

These issues could be resolved via a form of automated and programmatic mechanism, enabling the exchange of requests (as well as advertisements of availability) for coverage and capacity between different entities, such as for example between one MNO and one or more SNOs to provide NTN coverage and capacity in reaction to changing demand or to an unforeseen event, such as a disaster scenario.

There are currently no existing interfaces, APIs or otherwise programmatic and/or automated mechanisms to support either static, semi-static or let alone dynamic requests for coverage and/or capacity between different operator entities within either the 3GPP or the O-RAN specifications. Such requirements and concepts have been raised and discussed from time to time, at least within 3GPP in the context of NTN, but have never so far been introduced. In particular, there is currently no satisfactory mechanism to satisfy certain mission-critical use cases for the public good, such as respond to the need to supply Disaster Coverage via NTN in response to total or partial outages to parts or totality of an MNO's terrestrial network within meaningfully responsive timeframes from the occurrence of an event (e.g. seconds or at most minutes).

5.1.5 Use Case Group 1: Summary, Proposed Solution, Benefits and Impact to the O-RAN Architecture

5.1.5.1 Summary of Gap Analysis

The previous clauses describe a number of use cases that involve interaction between the TN and NTN networks and summarize the current gaps and required interactions. Due to the commonalities between the different use cases, it was decided to address possible solutions in a single, common clause. The following clause thus analyzes possible common solutions to address the aforementioned use cases. From analyzing the different use cases in Group 1, we can also summarize a list of requirements that a possible solution must be able to fulfil.

In all the previously discussed use cases of Group 1, it is assumed that the TN and NTN segments could be either part of the same operator domain or, more typically, part of two different and independent operator domains. It is also possible that, within a single operator domain, the operator may choose to maintain domain separation between the TN and NTN segments. Any new solution(s) must be capable of addressing all such cases.

In addition, the solution should not be limited to a one-to-one relationship between TN and NTN segments, but should also enable one-to-many relationships, for example to support the case where an MNO may have multiple SNO partners, and vice-versa the case where an SNO may have multiple MNO partners/customers, but also the case where multiple separate TN and NTN segments exist within a single operator, as separate administrative domains.

The key requirements for the solution can thus be summarized as follows:

- 1) Allow interaction between TN and NTN segments in the same operator, assigned to a single administrative domain
- 2) Allow interaction between TN and NTN segments in the same operator, assigned to multiple separate administrative domains
- 3) Allow interaction between TN and NTN segments in different operators and independent operator administrative domains

- 4) Allow interaction between one TN segment and multiple NTN segments (in the same or different operator domains)
- 5) Allow interaction between one NTN segment and multiple TN segments (in the same or different operator domains)
- 6) Allow sharing of NTN radio cell coverage information from the NTN to the TN
- 7) Allow sharing of TN radio cell coverage information from the TN to the NTN
- 8) Allow sharing of cell configuration between the TN and NTN
- 9) Allow sharing of NTN satellite assistance information with the TN (e.g. in the form of satellite Ephemeris for SIB19/SIB31)
- 10) Allow secure sharing of information between different operator trust domains
- 11) Allow dynamically-selective sharing of information across TN and NTN based on respective operator's control
- 12) Allow advertisement of network capabilities
- 13) Allow requesting of NTN coverage to one or more target locations in 3D space and time
- 14) Allow requesting of NTN capacity to one or more existing target NTN coverage areas
- 15) Allow sharing of load information between networks
- 16) Allow sharing of UE capability information between networks
- 17) Allow sharing of QoS and Slice policy information between networks, including for specific UE to be prioritized (e.g. list of UEs)
- 18) Allow sharing of information necessary to configuration of measurements for mobility between TN and NTN (in both directions)
- 19) Allow triggering of measurements from the partner network (TN or NTN), e.g. in preparation for a mobility event
- 20) Allow triggering of mobility events for Idle and Connected mode mobility between TN and NTN (in both directions)
- 21) Allow sharing of acknowledgements and status information for previously advertised or requested capabilities

The requirements described above cannot be presently satisfied by any existing interface, architectural element or procedure in the current 3GPP or O-RAN Architecture, at least for the case where the TN and NTN segment are operated by different entities (e.g. different operators) or part of different administrative domains.

The O-RAN architecture and RIC/SMO complex, however, do offer a unique framework for management of radio coverage in an operator domain, to complement the 3GPP specification and address challenges not addressed by 3GPP, and are either already capable, or are expected to be capable (e.g. in the case of NTN) of accessing the required information in the respective operator networks. They therefore offer a unique opportunity to introduce a viable solution for TN-NTN radio network coverage level inter-working and cooperation, if a suitable mechanism to enable communication between the TN and NTN domains can be identified.

5.1.5.2 Proposed Solution: New TN-NTN Interface

In order to satisfy the requirements outlined above, for the scenario in which the TN and NTN segments are operated by different operators, an MNO and an SNO respectively, or under otherwise different administrative domains, a potential proposed solution is the introduction of a new and novel O-RAN interface or API between the two operator domains that is suitably designed to cross the inter-operator domain trust boundary.

If we assume that both the TN and NTN are respectively adopting the O-RAN architecture and thus both have their own respective Near/Non-RT RIC and SMO deployments, the new interface could be designed to exist between the MNO's and SNO's respective Near/Non-RT RICs or SMOs, facilitating different levels of information sharing. Such capability and interface are not currently specified or contemplated by the O-RAN Architecture and Specifications, and therefore it is identified as a gap. Overall, this cross-operator mode of operation and interaction was not envisaged by 3GPP or the O-RAN Alliance previously and would require investigation into how the O-RAN architecture may need to evolve, potentially with the specification of one or more new interfaces.

After several discussions on at which layer should this new interface exist, and considerations for various candidate options, it seems that the most likely suitable approach may be to specify this interface between the Non-RT RICs or SMOs. However, the possibility of supporting a new interface between the Near-RT RICs could also be considered and should be further studied if the requirements for use cases demand it.

5.1.5.3 Impact to the O-RAN Architecture

This clause of the Technical Report outlines different architecture options for TN-NTN interworking depending on whether the two segments are operated by the same or a different operator (or different administrative entities within the same operator), and considers as a solution the specification of at least a new interface between an NTN Non-RT RIC or SMO and a TN Non-RT RIC or SMO, at least for the case where TN and NTN segment are operated by different entities.

The impact can thus be summarized as follows:

- **Case 1 (TN and NTN operated by different entities):** Introduction of a new East/West API interface or service between Non-RT RICs or SMOs of the two operators.
- **Case 2 (TN and NTN operated by the same entity):** Reuse of the existing A1 interface reference point and augmentation of the A1 and the RIC/SMO layer with procedures and information relating to TN-NTN information sharing.

For the scenario in which the TN and NTN segment are operated by different operators, an MNO and an SNO respectively, or part of different administrative domains, a potential proposed solution is to introduce a new East/West API interface or service between the SNO's NTN Non-RT RIC and the MNO's TN Non-RT RIC. Via such interface, the TN Non-RT RIC and the NTN Non-RT RIC could exchange information regarding the respective coverage areas as well as other information as required for mobility and service continuity procedures.

Figure 5.1.5.3-1, below, depicts the aforementioned topology in an inter-operator scenario, which sees separate TN and NTN network segments operated by different operators (e.g. an MNO and an SNO respectively), and highlights the potential new interface reference point between Non-RT RICs of the two operators. It is important to note that today some control and management functions may still be supported outside of the O-RAN architecture framework as they are not currently covered by any O-RAN function and are opaque to the O-RAN architecture. These have been represented in the figure below as “Non-O-RAN NTN Functions” and are currently outside of the scope of O-RAN and of this Technical Report. Similarly then, the interface between the Non-O-RAN NTN Functions and NTN gNB is currently a bespoke interface, which is also outside of the scope of O-RAN and of this Technical Report.

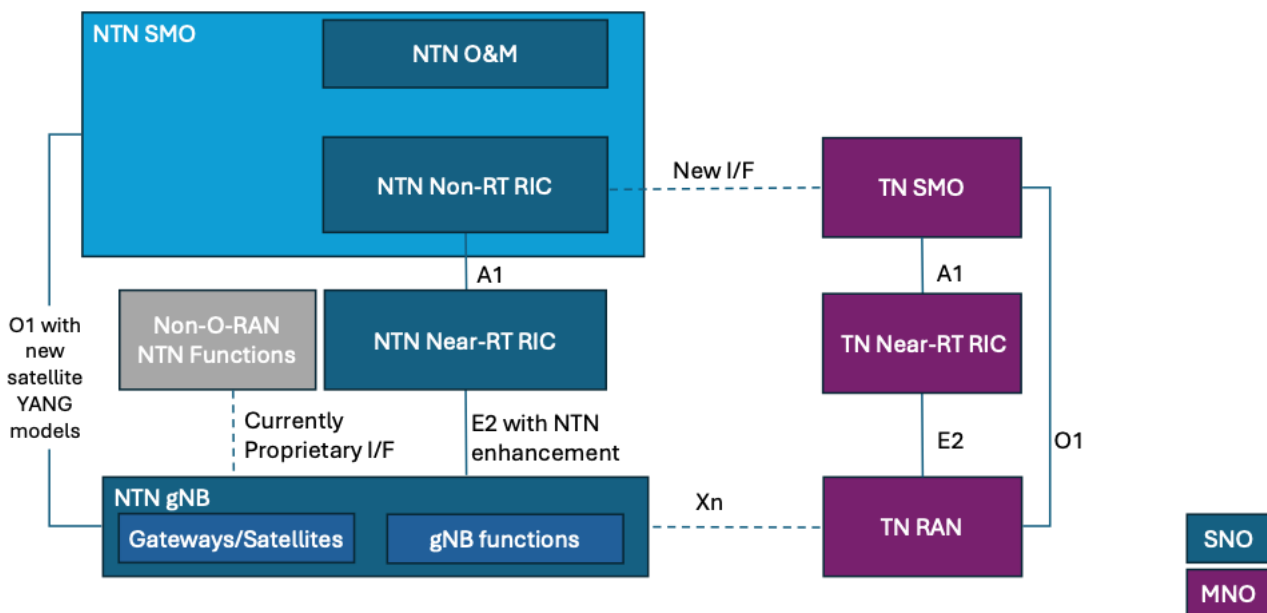


Figure 5.1.5.3-1: Topology of inter-operator scenario

For the case in which the TN and NTN segment are operated by the same operating entity, the existing architecture and interfaces could be reused, whereby a single Non-RT RIC and SMO may exist and assume the responsibility of managing both the TN and NTN segments of the network.

The existing mechanisms of interaction between the Non-RT RIC and the Near-RT RIC via the A1 interface could in principle thus be reused, with the assumption that the operator's Non-RT RIC would have multiple A1 interface associations with one or more TN Near-RT RICs, as well as one or more NTN Near-RT RICs. The common Non-RT RIC could then supply information to both the TN and NTN Near-RT RICs in the shape of A1 policies or enrichment information, with appropriate extensions to the standardized data types, yet to be determined.

Figure 5.1.5.3-2, below, depicts the aforementioned topology where both TN and NTN segments are operated by the same operator (e.g. either an MNO with an NTN component, or an SNO with a TN component). The topology of both TN and NTN segments follows the existing O-RAN architecture and reuses the existing O-RAN interfaces, such as the A1 reference point. Similarly to the inter-operator case, it is expected that today some control and management functions may still be supported outside of the O-RAN architecture framework as they are not currently covered by any O-RAN function and are opaque to the O-RAN architecture. These have been represented in the figure below as “Non-O-RAN NTN Functions” and are currently outside of the scope of O-RAN and of this Technical Report. Similarly then, the interface between the Non-O-RAN NTN Functions and NTN gNB is currently a bespoke interface, which is also outside of the scope of O-RAN and of this Technical Report.

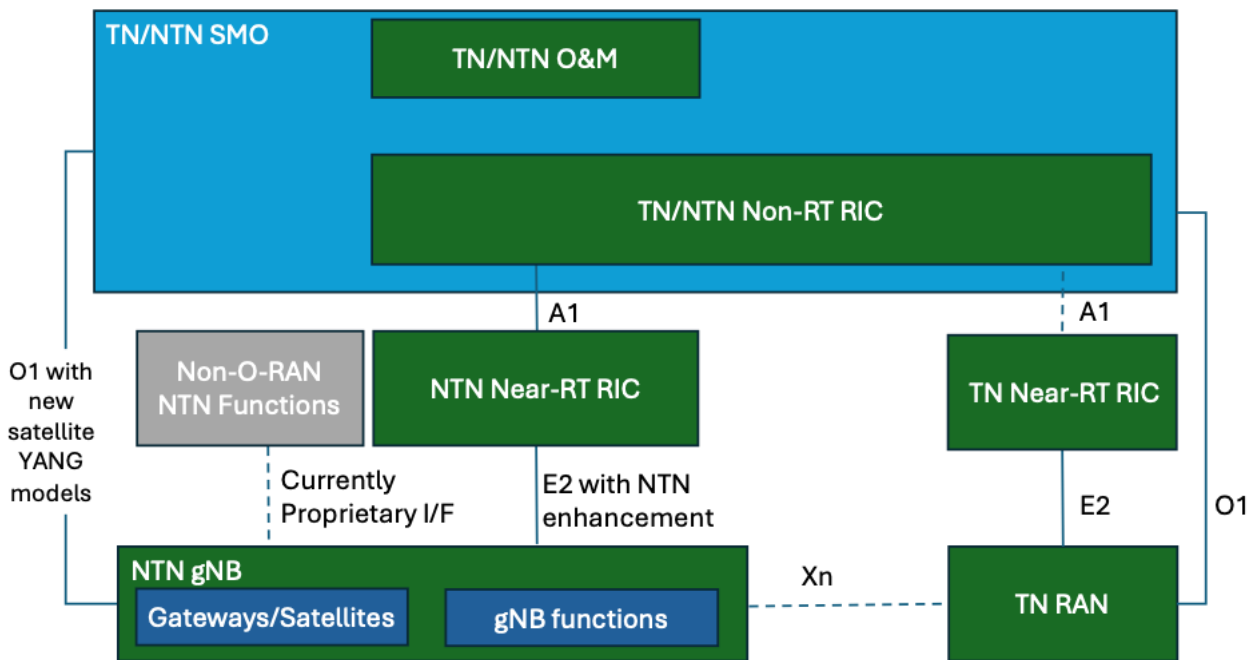


Figure 5.1.5.3-2: Topology of single operator scenario

5.1.5.4 Benefits

There are several benefits to this new proposed interaction, some of which have been captured in the Use Cases of the Use Case Group 1, and include the enablement of TN to NTN and NTN to TN mobility and service continuity scenarios, such as reselection and handover between the TN and the NTN and vice-versa, the enablement of the ability to request NTN coverage in a number of operator scenarios, as well as opportunities for higher levels of optimization, for example in how coverage is provided between the TN and NTN segment, and in ensuring truly complementary and mutually-beneficial and optimal coverage between the TN and NTN segments.

5.2 Use Case Group 2: Intra-NTN Mobility Use Cases

5.2.1 Overview

This clause describes use cases applicable within the NTN, in particular, use cases related to aspects of mobility and intra-system service continuity. Mobility events in an NTN constellation may be driven by a number of factors and impairments that are specific to NTNs, in addition to typical UE mobility requirements, such as satellite (i.e. network node) mobility as well as delay and Doppler, spectrum and power constraints and others such as line-of-sight. The following are some examples of key mobility scenarios that can be expected in an NTN system:

- Feeder link management (e.g. switching, load balancing, etc)
- Satellite mobility
- Satellite load balancing and selection
- UE Mobility
- Inter-satellite Link (ISL) mobility

3GPP Releases 17, 18 and 19 introduced new procedures specifically to address NR-NTN and IoT NTN mobility management, that are designed to take into account such NTN-specific impairments and influencing factors. However, whilst the procedures are specified in detail in the 3GPP specification, their actual implementation and the full implications from a network architecture and implementation are not stipulated and are instead left up to operator and/or vendor implementation choice.

This chapter analyses a number of mobility use cases that fall in the categories listed above and that leverage existing 3GPP procedures as of Release 18, within the context of the O-RAN architecture as a possible implementation, in a bid to efficiently allocate the limited resources of the NTN system whilst maintaining and maximizing fulfilment of user service requirements.

Lastly, an initial assessment of the possible impact of the introduction of such use cases to the O-RAN architecture and specifications is made.

5.2.2 Use Case 2-1: Connected mode NTN inter-gNB handover (HO) due to Satellite mobility

5.2.2.1 Use Case Description

This use case covers the use of an NTN near-RT RIC and xApps to optimize the implementation of inter-satellite inter-gNB Connected Mode Handover (HO) within an NTN Non-Geostationary Orbit (NGSO) constellation, exemplified by a Low Earth Orbit (LEO) constellation, driven by satellite mobility. This use case is in principle formulated in a way that is largely agnostic of the NTN gNB architecture (e.g. Transparent vs Regenerative). In this scenario, users are assumed to be practically stationary or slow moving (e.g. handheld), and the mobility events are driven by satellite motion. This scenario use case description assumes Earth-fixed cells (EFC) or quasi-Earth-fixed cells (QEFC), but is applicable also to Earth-moving cells (EMC).

Many upcoming 5G NTN constellations will be deployed in Low Earth Orbit, where a significant number of satellites are needed in order to achieve ubiquitous global coverage. LEO constellations are highly-mobile, and as such UEs may experience frequent handovers from one satellite to the next.

In addition, in dense constellations, multiple candidate satellites may be able to serve a given coverage area, introducing the need for the network to select the appropriate target satellite to serve the incoming users. This could be influenced by a number of factors, such as feeder link availability, expected satellite visibility period, maximum number of concurrent beams, on-board power limitations, UE TX power and antenna gain, line of sight, spectrum constraints and more.

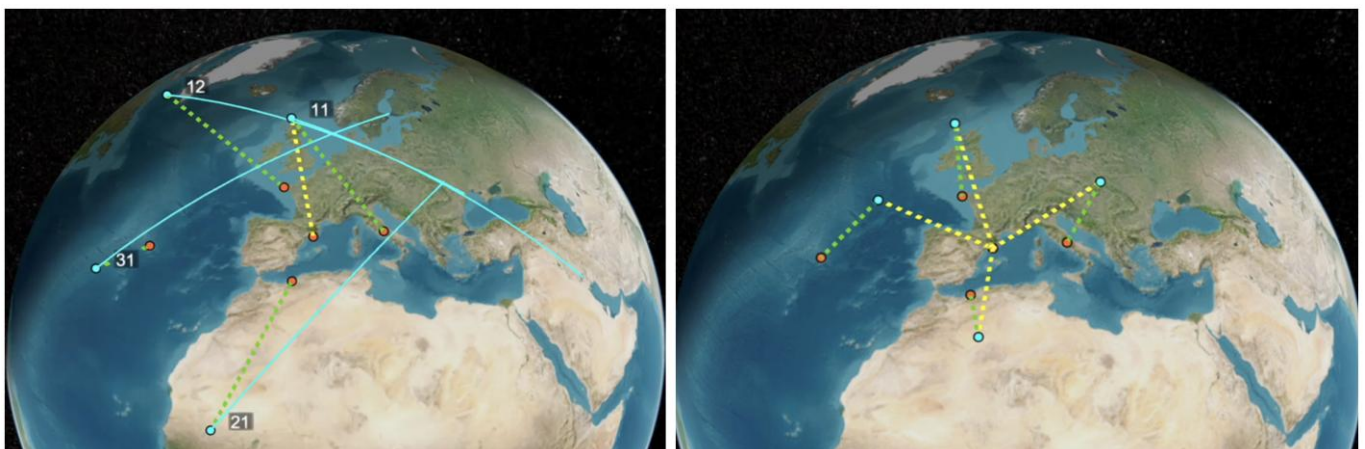


Figure 5.2.2.1-1: Example network scenario with 4 satellites from a Walker Delta formation (left) and snapshot of the moment where HO opportunity is available with 4 satellites in view (right)

The example scenario in Figure 5.2.2.1-1 shows part of an NTN LEO Walker Delta constellation with four satellites in view, two of which co-planar and two cross-planar. The projected satellite orbital trajectories are shown in cyan. As a simplifying

assumption, each satellite is associated with a single feeder link (shown in dotted green) to a given gateway (represented by red dots associated to the green dotted lines) and a single gNB broadcasting a single PCI. Feeder link mobility is thus not considered for the sake of simplicity. This allows this use case to focus on user link mobility, and makes the scenario applicable to both Transparent and Regenerative architectures.

Co-planar satellites are numbered with the same most-significant digit, whereas cross-planar satellites are assigned a different most-significant digit.

A UE located in Spain is currently served by satellite 11, which forms the user link to the UE (in yellow) and is supported by a feeder link (in green) to a ground station in central Italy. As shown by the orbital trajectory, in some time, satellite 11 will be approaching the end of its viable service time for the target UE (e.g. due to low elevation, scan angle or link budget). Simultaneously, the remaining three satellites 12, 21 and 31 will approach the UE coverage area and become eligible to place a beam (and a cell) in the UE's location, and they therefore offer potentially viable candidate handover opportunities. Figure 5.2.2.1-1 shows the possible candidate satellites and UE service links at the instant in which all satellites are in principle eligible to serve the coverage area.

An NTN Near-RT RIC and an associated NTN xApp could be tasked with taking all the various factors into account and driving the decision of which of the 3 available candidate handover satellites should start serving the UE once its currently serving satellite becomes unsuitable.

In addition, the previously mentioned power and spectrum constraints present a further optimization opportunity to the network and xApp: for example one of the available satellites could deliver the requested level of service to the UE while consuming fewer resources than another satellite. Lastly, in a scenario with several UEs in a given coverage area, the RIC and xApp must make a handover decision that accounts for the requirements and conditions of all the users in the coverage area, which may create the need for trade-off decisions.

5.2.2.2 Problem statement and challenges

The first part of the problem statement is relatively trivial. In order to support NR NTN scenarios aligned with 3GPP Release 17 and above, the Near-RT RIC and xApp must be able to configure the required NTN-specific measurement configurations, as well as the necessary handover triggers.

In Release 17, a new SSB Measurement Timing Configuration (SMTC) specific to NR NTN, namely SMTC4 (*SSB-MTC4List-r17*), has been introduced. In addition, the NTN cells must be able to broadcast Satellite Assistance information, in particular Satellite Ephemeris, in the form of SIB19, containing serving and neighbor satellite and cell information. Furthermore, in order to trigger a HO, the gNB must be able to include specific SIB19 content in the HO triggers.

This implicitly requires the NTN Near-RT RIC and xApp(s) to be aware of the constellation Ephemeris, the serving and target satellite timing information, as well have the ability to deliver the necessary configurations to the O-CU/DU.

The second aspect of the problem is related to coordination of the HO across multiple satellites (and cells, potentially gNBs), in order to hand the UE over to the best satellite.

Several criteria and KPIs may be taken into account to select the best target satellite, and, depending on the service requirements set by the SNO, the NTN RIC may have to either privilege one or more specific KPIs at the expense of others, or to perform a multi-constraint evaluation by ensuring adequate and balanced satisfaction of several different KPIs at once.

In terrestrial networks, assessment of handover opportunities is almost entirely focused on KPIs affecting the user link, and would typically be driven by UE measuring the SSBs of relevant neighbour cells, and then the gNB or Near-RT RIC monitoring for SMTC-configured measurement events where the quality on a neighbour cell signal would exceed a given threshold, while that of the serving cell would drop below another threshold, such as is the case in the A3 event. Measurement reporting, based on SSBs but also CSI-RS, is also very important for channel estimation in terrestrial networks to aid link adaptation, primarily due to urban deployments where the various channel state and effects of multi-path propagation are highly and frequently variable and difficult to predict, except perhaps at the cell edge. In NTN deployments, on the other hand, limited multipath effects due to focus on outdoor environments, and the predictable movement of the satellite in the sky coupled with accurate weather forecast data make it such that the user link channel state by itself is much more steady and predictable, driven mostly by satellite motion, and can thus be quite reliably modelled and estimated. The overall link and HO candidate selection is, however, on the other hand, affected by a number of other factors, many of which are completely opaque to the gNB and cannot be addressed with measurements.

Some examples include:

- Ability to sustain the link vector for the duration of the satellite pass (or UE call flow) - influenced by a number of factors including feeder link availability
- Available on-board battery power
- Overall network load balancing goals
- Cleanest channel, and thus best achievable MCS
- Feeder link availability - e.g. due to weather or other system constraints such as antenna field of regard and LoS to the satellite.
- Line of sight (LoS) on the user link (but not only)

Feeder link availability is a particularly interesting example as it is chiefly important in transparent NTN deployments where, even in FR1 user link systems, the NR-Uu waveform has to traverse also the feeder link between the satellite and the ground station, which often utilises higher frequencies (e.g Ku, Ka-, Q- or V-band) that are notably susceptible to weather attenuation, and which makes it such that the feeder link directly impacts the NR-Uu user link radio interface. The feeder link behavior is not specified in 3GPP, this effectively makes it opaque to the gNB. In addition, due to the dynamic nature of the system and of the external effects (e.g. weather) that might intervene, the impact to the NR-Uu user link introduced by the feeder link might not be completely known in advance by any of the individual gNBs at the moment of handover, due to the fact that a feeder link might not even have been established yet.

Continuous satellite movement also makes it such that the channel quality is, albeit predictably, always continually and steadily changing, driven effectively by gNB motion, unlike in terrestrial networks where, if the UE is stationary, the base station is in most cases static.

Not only might the different satellites and their respective feeder and user link vectors, propagating through variable weather and atmospheric conditions, result in varying channel qualities and therefore attainable MCS regimes, but they can also result in varying viable service times before either the user link or the service link becomes unsustainable, due to factors such as Line-of-Sight (LoS) or minimum elevation angles due to field-of-view (FoV) and obstructions.

Thus, serving or neighbour cell signal measurements prior to the handover can by themselves be at least partially misleading in NTN, since they do not indicate much about either the end-to-end channel quality including the feeder link and the satellite itself (e.g. on transparent systems), channel state evolution throughout the satellite pass, or future vectors between each satellite and the UEs being served, or expected to be served in the future (e.g. impacted by LoS effects, such as mountain ranges which might block LoS from a given area and UEs or feeder links to a given satellite at any future point of the service pass). For example, for Satellite 31 in Figure 5.2.2.1-1, a feeder link is formed with a ground station in the Azores. As the satellite propagates due North-East, it may witness its current ground station fall out of view before the UE does, thus requiring a feeder link handover. While feeder link is not a 3GPP-specified element of the network, mobility procedures associated with it may still require intervention from the RAN, and, depending on network implementation, may potentially result in an L3 Handover as well for the UE, leading to potential service discontinuity and signaling overhead.

Furthermore, since the NTN link is often more constrained compared to a terrestrial link, an excessive reliance on user link signalling and UE measurement reports may create significant overhead on the link, negatively affecting user throughput and system capacity.

This introduces the need for intervention, and opportunity for optimization, from an external control plane that has awareness of the satellite constellation, its sub-components and its evolution, and is thus capable of predicting the behaviour of both the user links and their channel state across multiple satellites throughout their motion, as well as the impact of components of the system that are outside of the scope of any individual gNB, such, in this example, other gNBs, other satellites or NTN service nodes, or any of the gateway feeder links. In addition, to account for events in the future impacting links that have not come into existence yet, there might be a need for the NTN control plane to be able to be spatially-aware, in addition to it being temporally-aware, to propagate predict where the NTN service nodes (e.g. satellites, but not only) might be at any future point in time and which links they might be able to form.

5.2.2.3 Proposed Solution

A possible solution could be leveraging the NTN RIC, either as a combination of the Near-RT and the Non-RT RIC, or via the Near-RT RIC alone, with the introduction of a new control loop provided by one or more NTN xApp(s). Such xApps would leverage the holistic view of the satellite network provided by the NTN RIC, combined with predictive capabilities, to produce handover decisions which take into account multiple concurring factors, as well as reducing the reliance on signal measurements and reporting.

The NTN RIC and xApp(s)/rApps are configured with a digital twin of the NTN network, including, not only the 3GPP RAN segment, but also the NTN ground and space segments (e.g. the gateways, the feeder links, the satellites, etc) via which the NTN

service is delivered, even though some of these elements are presently not within the scope or control of the existing O-RAN reference framework.

It is assumed that the NTN RIC may be capable of predicting NTN network node state, motion, trajectory (e.g. orbit) and behaviour in time and space. It is also assumed that the NTN RIC may be able to ingest external data sources, such as, but not limited to, weather data, to predict impact of external events on the active or candidate links, in order to aid the control loop affecting handover decisions.

By knowing the predictable orbital movement of the candidate satellites , their association with ground stations and feeder links, the relatively static position (compared to the satellites) of UE or groups of UE in a given target coverage area (e.g. cell or cell portion) and the impact of weather effects, an NTN xApp could drive the decision of which satellite to hand over the UE to by modelling the candidate link vectors which could be formed and sustained until the need for the next handover, and aiming to choose the one that satisfies the appropriate combination of constraints. The use of prediction can be considered a supplement to, not a replacement of the evaluation of UE measurement reports.

Table 5.2.2.3-1: NTN handover management

Use Case Stage	Detailed Information
Goal	Find best handover candidate satellites/cells for a UE to address impact of satellite motion and constellation evolution, while considering a number of diverse constraints and dependencies not directly visible through measurements, and trigger a Connected Mode Handover (HO).
Actors and Roles	NTN xApp, Near-RT RIC, E2 Nodes, UE.
Assumptions	All relevant functions and components are instantiated. The NTN RIC is aware of the state and motion of the constellation and is capable of predicting its evolution in time and space.
Pre-conditions	E2 interface connectivity is established, UE is in active RRC Connected Mode state to a given serving satellite (e.g. Satellite 11)
Step 1 (M)	NTN Handover Optimization xApp/Near-RT RIC subscribes to UE information change and message copy of UE mobility from E2 Nodes to understand the state of the network and the UEs within it.
Step 2 (M)	NTN Handover Optimization xApp/Near-RT RIC requests to the E2 Nodes for the UEs to perform measurements of both serving and visible candidate target cells across satellites, and configure in the E2 Nodes UE information change and message copy of UE mobility to be reported back to the NTN Handover Optimization xApp /Near-RT RIC.
Step 3 (M)	E2 Nodes configure target measurements to UEs and UEs perform measurements and monitor for configured measurement events, upon which they signal a measurement report to the serving NTN E2 Node
Step 4 (M)	NTN E2 Nodes forward the measurement report to NTN Handover Optimization xApp/Near-RT RIC
Step 5 (M)	xApp with digital twin analyses the currently employed Ground Station and transparent LEO satellite to derive the remaining service time before a handover must occur. It also evaluates the candidate handover opportunities via available neighbouring satellites and feeder links to model the link performances attainable during the service times that they could respectively offer to the UE, and compares them with the results of the measurement reports. Its output is a selection of the neighbour satellite/cell which is best positioned to serve the UEs in the target coverage area.
Step 6 (M)	NTN Handover Optimization xApp/Near-RT RIC signals to NTN E2 Node to establish the correct handover control to the target UEs, to the selected target satellites, including the correct SIB19 contents (e.g. Ephemeris) for the respective target satellites to be included in the HO request message by the E2 Node.
Step 7 (M)	NTN E2 Node implements RRC Connected Mode handover as specified by NTN Handover Optimization xApp/Near-RT RIC

```

@startuml handover optimization
title NTN Handover Optimization Use Case
Skin rose
skinparam sequenceMessageAlign left
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam defaultFontSize 12

box "NTN Segment" #cyan
participant "NTN HO Optimization xApp/Near-RT RIC" as xappnearrrtric
participant "NTN E2 Nodes" as e2nodes
end box

autonumber

group #White NTN Handover Optimization xApp/Near-RT RIC collect RAN metrics
xappnearrrtric -> e2nodes: <E2> Subscribe to UE information change and message copy of measurement reports
e2nodes -> xappnearrrtric: <E2> Report UE information change and message copy of measurements events
end group

group #White NTN Handover Optimization xApp/Near-RT RIC establish signal measurement and reporting config for candidate handover opportunities
xappnearrrtric -> e2nodes: <E2> Set up signal measurement and reporting configuration per-UE/per-cell
note over e2nodes: Configure measurement control for candidate serving satellites/cells and\nUE report measurement result of candidate serving satellites/cells to E2 nodes
e2nodes -> xappnearrrtric: <E2> Report the measurement result
end group

note over xappnearrrtric
xApp with digital twin analyses the currently employed Ground Station and transparent LEO satellite to derive the remaining service time before a handover must occur. It also evaluates the candidate handover opportunities via available neighbouring satellites and feeder links to model the link performances attainable during the service times that they could respectively offer to the UE, and compares them with the results of the measurement reports. Its output is a selection of the neighbour satellite/cell which is best positioned to serve the UEs in the target coverage area.
end note

note over xappnearrrtric : xApp derives decision to handover\nUE to Neighboring X LEO, served\nby Neighboring X gNB/NTN GW

xappnearrrtric -> e2nodes: <E2> Per-UE network-initiated HO configuration, inclusive of SIB19 satellite assistance information
note over e2nodes: Network initiated handover

@enduml

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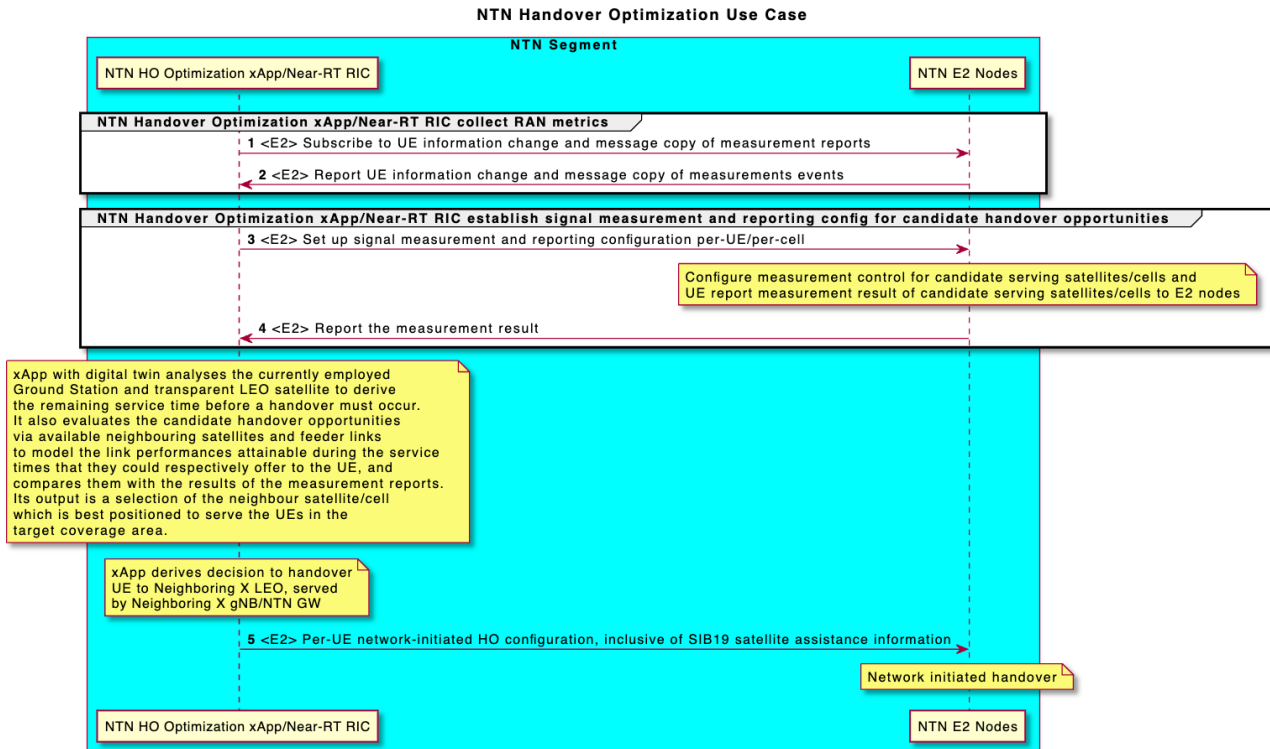



Figure 5.2.2.3-1: UML chart of NTN handover management

5.2.2.4 Benefits

The first and main benefit is the introduction of support of Release 17+ NR NTN network-controlled Handover (HO) from RRC Connected Mode, including support for NTN-specific satellite assistance information in the HO request/trigger, which is presently not supported by the O-RAN specifications.

The second benefit, achieved by the introduction of an NTN RIC (Near/Non-RT RIC) that is aware of the dynamic topology, physics and evolution of the constellation, and is capable of predicting its behaviour, is a potentially significant reduction of the signalling overhead, as well as a reduction of the potential for handover failures or sub-optimal handovers, leading to an improved service experience, as well as overall improved efficiency of the network by reduction of the resources consumed to support a mobility event.

5.2.2.5 Impact on O-RAN WGs

The use case brings an intelligent NTN handover management approach to O-RAN, which also utilizes information that is not natively exposed by the O-RAN interfaces (information owing to the non-3GPP components of an NTN constellation, and the constellation's own dynamic evolution).

There is a need to support the configuration of the previously described new measurement configurations specific to NR NTN introduced in Release 17, such as the new SSB Measurement Timing Configuration (SMTTC) or SMTTC4 (*SSB-MTTC4List-r17*). In addition, the NTN cells must be able to broadcast Satellite Assistance information, in particular Satellite Ephemeris, in the form of SIB19, containing serving and neighbor satellite and cell information. Furthermore, in order to trigger a HO, the gNB must be able to include specific SIB19 content in the HO triggers.

These could be implemented by extending the existing support for measConfig and configuration of HO that is found in E2 Service Model RAN Control, as well as investigating new mechanisms either via E2 or O1 to deliver SIB19 information to gNB from the RIC/SMO.

5.2.3 Use Case 2-2: NTN QoS-based inter-cell traffic steering and load balancing

5.2.3.1 Use Case description

Void.

5.2.3.2 Problem statement and challenges

Void.

5.2.3.3 Proposed Solution

Void.

5.2.3.4 Benefits

Void.

5.2.3.5 Impact on O-RAN WGs

Void.

5.2.4 Use Case 2-3: NTN Time-based and Distance-based Conditional Handover (CHO)

5.2.4.1 Use Case Description

Void.

5.2.4.3 Problem Statement and challenges

Void.

5.2.4.4 Proposed Solution

Void.

5.2.4.5 Benefits

Void.

5.2.4.6 Impact on O-RAN WGs

Void.

5.2.5 Use Case 2-4: NTN Idle mode cell re-selection

5.2.5.1 Use Case Description

Void.

5.2.5.3 Problem Statement and challenges

Void.

5.2.5.4 Proposed Solution

Void.

5.2.5.5 Benefits

Void.

5.2.5.6 Impact on O-RAN WGs

Void.

5.2.6 Use Case 2-5: NTN UE Location-aware Connected Mode Handover and Inter-cell Traffic Steering and Load-balancing

5.2.6.1 Use Case description

Void.

5.2.6.2 Problem statement and challenges

Void.

5.2.6.3 Proposed Solution

Void.

5.2.6.4 Benefits

Void.

5.2.6.5 Impact on O-RAN WGs

Void.

6 Summary and Conclusion

Void

Annex:

Change history/Change request (history)

Date	Revision	Description
2024.12.12	00.00.03	Initial TR skeleton
2025.02.24	00.00.04	- Add introduction and NTN development status in standard organizations - The following CRs have been implemented: - KEY, AAL - CR 0001 – Introduction - v2 - KEY, AAL, FRN – CR 0002 - NTN development status in standard organizations - v3
2025.06.17	00.00.05	- Introduction of initial Use Case Group 2 set - The following CRs have been implemented: - AAL, KEY, RAD – CR 0004 - Introduction of Use Case Group 2 - v0.6_NOK_AAL
2025.06.18	00.00.06	- Update clause 1-3 and change history - introduction of Use Case Group 1, the following CRs have been implemented: - KEY, AAL – CR 0005 - fix clause 1-3 and change history - v3 - AAL, KEY, RMI, MTR – CR 0003 - Introduction of Use Case Group 1- v0.12_AtMeeting
2025.06.30	00.00.08	- Updates to the overall TR - Corrections to image references after first round of comments.
2025.07.09	00.00.09	Editorial and formatting for first release
2025.07.21	00.00.10	- Implementation of voting round editorial comments - Addition of CR references to the Change history/Change request (history)
2025.07.22	01.00	First formal release